

# UNSW-NB15 MAQT Training Setups

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This note summarizes how lawun chose MAQT training settings on FROZEN UNSW-NB15, what the loss curves mean, and what is still open. UNSW is harder: **8 known classes**, heavy Normal imbalance, and a smaller quantum width (PCA → 4 qubits vs 5 on BoT-IoT).

**Latest (full known train):** old inter loss (no margin / no hardest-only); cosine LR start 0.01;  $\lambda_1 = 0.66$ ,  $\lambda_2 = 0.33$ ; weighted sampler; patience 5. CE full-test still **~46%**. See §5.

Unless noted, “~46%” = CE-head on full known test, **no** conformal reject mask.

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## 0. Loss (two inter variants)

$$L = L_{CE} + \lambda_1 L_{intra} + \lambda_2 L_{inter}$$

| Term        | Role                                  |
|-------------|---------------------------------------|
| $L_{CE}$    | classify from circuit expectations    |
| $L_{intra}$ | pull $\rho(x)$ toward own prototype   |
| $L_{inter}$ | push away from wrong-class prototypes |

**Old inter** (`scripts/loss.py`, current train): mean negative TD to other-class prototypes. No margin, no hardest-only.  $L_{inter}$  is typically **negative**.

**Hinge inter** (v2.4 suggestion, tried then dropped): `relu(m - d)` on nearest wrong prototype (`hardest_only=True`). Margin  $m$  unpaid while  $d < m$ .

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## 1. Why BoT knobs don't copy

|                       | BoT-IoT | UNSW-NB15  |
|-----------------------|---------|------------|
| Known classes         | 4       | 8          |
| Full train            | ~148k   | ~81k       |
| Qubits (PCA)          | 5       | 4          |
| Selected CE full-test | ~68%    | stuck ~46% |

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## 2. First full run: weak $\lambda_2$ , hinge margin 0.6 → ~46%

**Setup:** full train; cosine 0.01;  $\lambda_1 = 0.5$ ;  $\lambda_2 = 0.01$ ; hinge margin 0.6; patience 5.

| Metric           | ~value                     |
|------------------|----------------------------|
| CE full-test acc | ~46% ( $n \approx 10086$ ) |

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| Metric                    | ~value |
|---------------------------|--------|
| CE full-test macro-F1     | ~0.26  |
| Algo-3 accepted-known acc | ~34%   |

$L_{CE}$  fell (LR 0.01 fine). Mean inter TD  $\sim 0.23\text{--}0.27 \ll 0.6 \rightarrow$  hinge unpaid;  $\lambda_2$  too small; ES cut  $\sim$  epoch 9.

### 3. Subset smoke: hinge $\lambda_2 = 0.1$ , margin 0.3

DPP cap 1000/class.  $L_{inter} \sim 0.05$ ; TD near 0.3. Looked promising  $\rightarrow$  promoted to full-data hinge trials.

### 4. Hinge on full data (margins 0.2–0.6) — no CE gain

Tried full train with hinge + hardest-only across margins 0.2–0.6 (incl. planned  $\lambda_2 = 0.1$ , margin 0.3, patience 15).

**Result:** CE full-test stayed **~46%**. Geometry knobs moved; accuracy did not.

$\rightarrow$  Dropped hinge. Reverted to **old inter loss**.

### 5. Latest: old inter loss (still ~46%)

**Setup:**

| Knob                   | Choice                                                                                                       |
|------------------------|--------------------------------------------------------------------------------------------------------------|
| Inter loss             | <b>old</b> (no margin, no hardest-only)                                                                      |
| Train                  | full FROZEN known                                                                                            |
| LR                     | cosine, start 0.01                                                                                           |
| $\lambda_1, \lambda_2$ | 0.66, 0.33                                                                                                   |
| Sampler                | weighted (keep Normal; don't hard-balance first)                                                             |
| Early stop             | patience 5                                                                                                   |
| Notebooks              | <a href="#">final-unsw-nb15-maqt-train</a> $\rightarrow$ <a href="#">final-unsw-nb15-algo3-cached (maqt)</a> |

Same ~46% CE full-test. Open: lift accuracy (retrain / other knobs), not “flip hinge back on” without a new signal.

### Decision trail

```
full / hinge /  $\lambda_2=0.01$  / margin=0.6 / patience=5
   $\rightarrow$  ~46% CE; hinge unpaid; ES short
```

```
subset / hinge /  $\lambda_2=0.1$  / margin=0.3  
  → geometry OK on smoke; not a CE win on full data  
  
full / hinge / margin  $\in \{0.2...0.6\}$   
  → still ~46% → abandon hinge + hardest-only  
  
latest: full / old inter /  $\lambda_1=0.66$  /  $\lambda_2=0.33$  / cosine 0.01  
  → still ~46% CE full-test
```